Robot Integration Team Project I
Lab 5 report: Perspective Geometry, Feature Matching and Lucas–Kanade Tracking

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***Abstract*— This report combines analytical camera geometry with an implemented feature-tracking pipeline for MIT VNAV 2023 Lab 5. Exact sphere silhouettes and vanishing points are derived for a pinhole camera. The experimental component implements SIFT descriptors, FLANN matching, a distance-ratio test, fundamental-matrix RANSAC, and Harris-initialized Lucas–Kanade (LK) tracking. The course image pair produces 101 ratio-filtered matches and 79 geometric inliers. Both specified ROS bags are processed completely using both algorithms: 1584 RealSense images and 1386 TESSE images, with exact source-timestamp agreement and no rejected images. Mean geometric inlier ratios are 95.73%/99.98% for SIFT/LK on RealSense and 86.06%/94.70% on TESSE. Sparse-correspondence failures inside OpenCV are explicitly recorded, not counted as correct matches. Supplementary descriptor and optical-flow tests are separated from the official-bag results.**

Index Terms—Perspective geometry, SIFT, FLANN, RANSAC, Harris corners, Lucas–Kanade tracking, ROS.

I. Introduction and Scope

Visual navigation requires reliable image correspondences before relative pose or triangulation can be estimated. Descriptor matching searches for similar local appearance, whereas differential tracking exploits temporal continuity and image gradients. Projection geometry explains the observations on which both operate.

The verified implementation, version `005`, preserves the course `FeatureTracker`, `SiftFeatureTracker`, and `LKFeatureTracker` interfaces and builds as ROS package `lab_5`. A headless entry point retains both course launch names and saves per-frame statistics. Earlier versions and shared environments are unchanged. Official-bag evidence is distinguished from supplementary EuRoC tests; the latter do not substitute for required recordings.

II. Perspective Geometry

A. On-axis sphere: exact silhouette

The optical center is the origin, the normalized image plane is $Z = 1$, and the camera has focal length one, zero principal point/skew, and unit pixel scale. For a sphere with radius r and center $C = (0,0,d)$, $d > r+1$, pixel (u,v) defines ray $P(\lambda) = \lambda(u,v,1)^T$, $\lambda > 0$. Substitution into $\|P-C\|^2 = r^2$ gives

$$(u^2+v^2+1)\lambda^2 - 2d\lambda + d^2 - r^2 = 0. \tag{1}$$

Silhouette rays are tangent, hence the discriminant is zero:

$$u^2+v^2 = \frac{r^2}{d^2-r^2}, \qquad \rho = \frac{r}{\sqrt{d^2-r^2}}. \tag{2}$$

The image is a filled disk bounded by this circle. The radius r/d describes a far-field approximation, not the exact silhouette: the sphere's tangent points do not all have depth d.

B. Off-axis sphere

Let $C = (a,b,c)^T$ and $S = a^2+b^2+c^2-r^2$. Tangency implies

$$(au+bv+c)^2 = S(u^2+v^2+1). \tag{3}$$

The expanded conic is

$$\begin{aligned}(b^2+c^2-r^2)u^2 - 2abuv + (a^2+c^2-r^2)v^2 \\ - 2acu - 2bcv + a^2+b^2-r^2 = 0.\end{aligned} \tag{4}$$

For an entirely front-facing sphere, $c > r$, its discriminant is $-4S(c^2-r^2) < 0$, so the silhouette is an ellipse. Its center

$$(u_0, v_0) = \left(\frac{ac}{c^2-r^2}, \frac{bc}{c^2-r^2}\right) \tag{5}$$

generally differs from $(a/c, b/c)$, the projected sphere center. The on-axis circle is a special ellipse. A sphere intersecting the camera plane or containing the optical center requires additional visibility analysis; a finite ellipse is not then guaranteed.

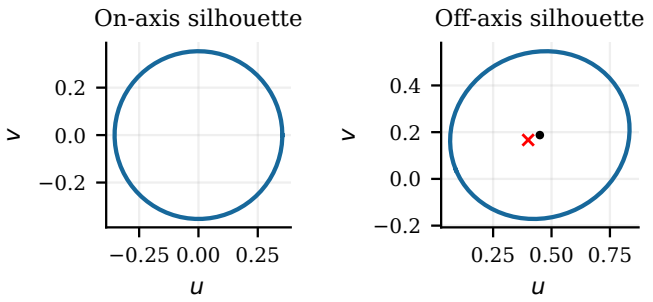


Fig. 1. Analytical examples, not measurements. Left: $r = 1, d = 3$. Right: $C = (1.2, 0.5, 3)$ and $r = 1$; black dot denotes the ellipse center, red cross the projected sphere center.

C. Vanishing points and parallel projections

Parallel spatial lines are $P_i + \lambda D$ with $D = (d_x, d_y, d_z)^T \neq 0$. If $d_z \neq 0$, taking the projection's infinite-distance limit gives

$$(u_\infty, v_\infty) = (d_x/d_z, d_y/d_z). \tag{6}$$

The starting point cancels, so all lines with this direction share one vanishing point. In homogeneous pixel coordinates, $v_\infty \sim KD$.

If $d_z = 0$, the vanishing point lies at infinity and the two finite image lines have the same direction (d_x, d_y). Conversely, distinct parallel image lines meet only at infinity, so their common spatial direction must have $d_z = 0$. This proves necessity and sufficiency after excluding coincident projections, optical-center lines that project to a point, and undefined camera-plane projections.

Numerical consistency tests gave tangency residuals of zero (on axis) and 1.07×10^{-14} (off axis), and a finite-distance vanishing-limit error of 2.81×10^{-8}. These checks do not replace the general proofs.

III. Feature Matching and Tracking

A. SIFT and FLANN

SIFT detects scale-space extrema and assigns a local orientation before computing gradient descriptors [2]. Each descriptor has 4×4 spatial cells and eight orientation bins, or **128 floating-point components**. Descriptor dimension is not image-coordinate dimension. Fig. 3 shows actual detections.

FLANN returns two approximate nearest neighbors for each descriptor. With distances $d_1 \leq d_2$, a match is accepted only if $d_1 < 0.8 d_2$. This removes ambiguous candidates whose best and second-best matches are similarly plausible. Float SIFT descriptors use Euclidean distance; binary ORB/BRIEF descriptors need a binary-compatible matcher rather than the same float configuration.

B. RANSAC and matching statistics

The course fundamental-matrix test uses RANSAC with a 3-pixel threshold and confidence 0.99. Homogeneous correspondences should satisfy $\tilde{x}_2^T F \tilde{x}_1 \simeq 0$. For accepted matches G_k and geometric inliers I_k in pair k, define

$$r_k = \begin{cases} I_k/G_k, & G_k > 0, \\ 0, & G_k = 0, \end{cases} \qquad \bar{r} = \frac{1}{N} \sum_{k=1}^{N} r_k. \tag{7}$$

The mean of ratios is not generally the ratio of aggregate counts. Invalid model estimates contribute zero inliers, not omitted rows. Geometric consistency is not correspondence ground truth; near-planarity and low parallax can make F poorly informative even with many inliers.

C. Harris and pyramidal LK

Harris corners are identified from the local gradient matrix

$$M = \sum_{x \in \mathscr{W}} w(x) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \quad R_H = \det M - 0.04 (\operatorname{tr} M)^2. \tag{8}$$

Two large eigenvalues constrain motion in both directions. Flat patches provide little information, while isolated edges exhibit aperture ambiguity. Under approximate brightness constancy and small, locally shared motion, linearization of $I(x + \delta x, t+1) \simeq I(x,t)$ gives a least-squares flow equation $M \delta x = -\sum w \nabla I I_t$ [4].

The implementation uses at most 500 Harris points, quality level 0.01, spacing 10 pixels, 21×21 LK windows, maximum pyramid level 3, and a 30-iteration/0.01 termination rule. It redetects when fewer than 50 tracks remain. Tracks must pass forward/backward status, finite-coordinate and image-bound tests, and a forward–backward discrepancy of at most one pixel. Pyramids increase capture range but do not solve occlusion or brightness changes. Redetection also means identities are not one fixed landmark set throughout the sequence.

TABLE I
COURSE BOX-PAIR C++ SIFT RESULTS

Metric	Value
Keypoints, image 1 / image 2	604 / 969
Descriptor dimension	128
Raw nearest-neighbor matches	604
Good matches after ratio filtering	101
Geometric inliers / outliers	79 / 22
Inlier ratio	78.2178%

IV. Experimental Protocol

The environment is ROS Noetic, OpenCV 4.13.0, and C++14; random seed 20261022 and single-thread OpenCV execution are used. The required recordings are `30fps_424x240_2018-10-01-18-35-06.bag` (RealSense) and `vnav-lab5-smooth-trajectory.bag` (TESSE). Their RGB streams contain 1584 images at 424×240 and 1386 images at 720×480. Processing converts RGB messages to grayscale.

Resumable upload produced identical Mac/server SHA256 checksums. The course box launch and four bag/algorithm launches were actually executed on private ROS masters. Playback was 0.2 times recorded speed to prevent offline processing from overflowing callback queues. This changes neither the original image content nor inter-frame displacement and is not a real-time benchmark.

For all four runs, processed header timestamps match the entire source sequence exactly: no missing, duplicated, or rejected image. Pair counts are 1583 and 1385, respectively. Figures and per-frame CSV files are saved alongside launch commands, parameters, checksums, and verification JSON.

V. Results and Discussion

A. Course image pair

Table I and Fig. 4 show substantial ambiguity reduction: 604 raw matches become 101 good matches, of which 79 satisfy the geometric test. The remaining 22 are model outliers. The approximately planar object geometry warrants treating this as a feature-matching demonstration, not a calibrated motion-accuracy benchmark.

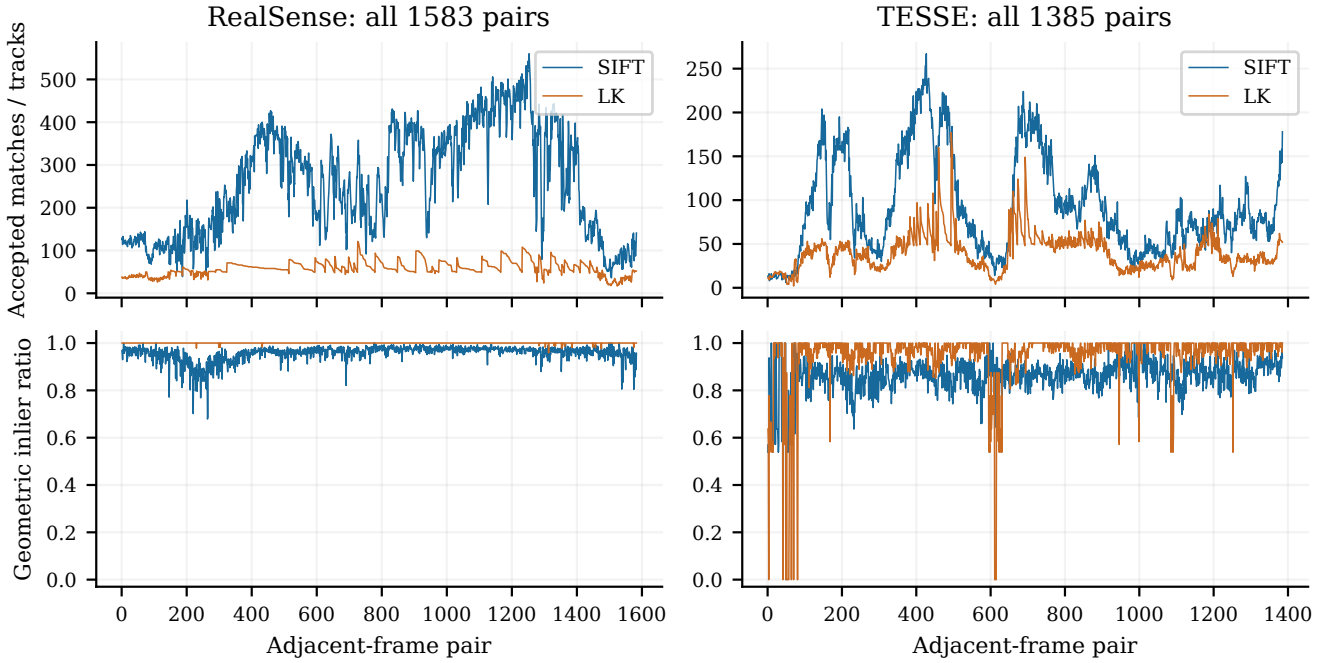


Fig. 2. Full experimental histories on the specified bags. Top: accepted descriptor matches or LK tracks. Bottom: geometric inlier ratios, including invalid model estimates as zero. Pair index is used rather than playback wall time; the protocol is described in Section IV.

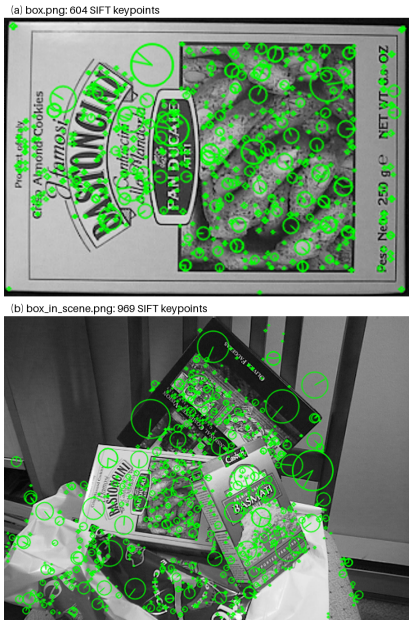


Fig. 3. Course box images with 604 and 969 SIFT detections. These are saved C++ outputs; marker size and orientation visualize the local feature frame.

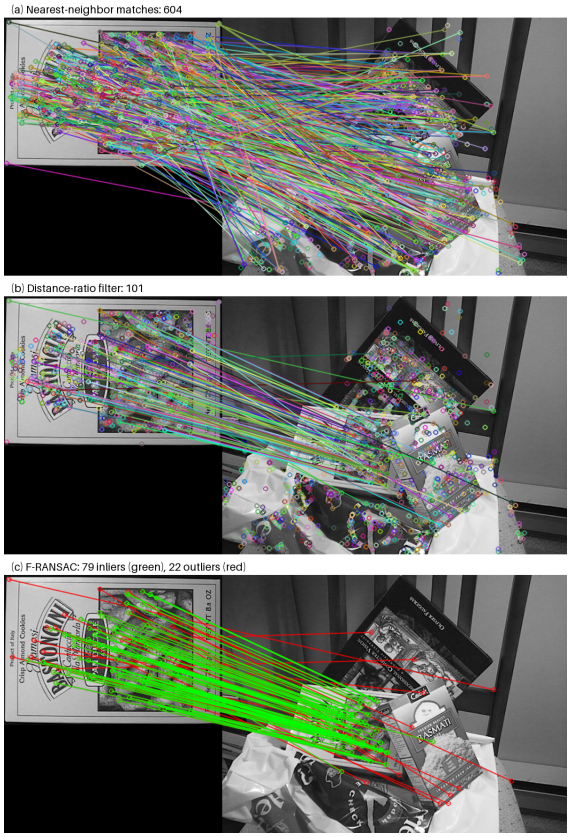


Fig. 4. Actual box-pair matches before ratio filtering, after the 0.8 test, and after F-RANSAC classification. The final panel shows model inliers in green and geometric outliers in red.

B. *Full official recordings*

SIFT produces more accepted matches than LK, whereas LK has a higher geometric inlier ratio in both recordings (Tables II and III). The latter is a smaller, more selective population after local-motion and forward–backward filtering; a higher ratio does not establish universally better tracking accuracy. Differences between recordings combine scene,

TABLE II

SIFT MEANS ON ALL OFFICIAL-BAG PAIRS

Metric	RealSense	TESSE
Previous-image keypoints	403.56	147.78
Current-image keypoints	403.62	147.93
Raw matches	403.56	147.78
Good matches	258.41	92.95
Geometric inliers	249.03	80.57
Mean inlier ratio	95.73%	86.06%
Images processed / expected	1584/1584	1386/1386

TABLE III

HARRIS+LK ON ALL OFFICIAL-BAG PAIRS

Metric	RealSense	TESSE
Mean previous-image corners	59.28	46.39
Mean accepted current tracks	58.45	40.19
Mean geometric inliers	58.44	38.83
Mean inlier ratio	99.98%	94.70%
Valid F estimates / pairs	1583/1583	1367/1385
Images processed / expected	1584/1584	1386/1386

texture, resolution, viewpoint, and motion effects rather than isolating one cause.

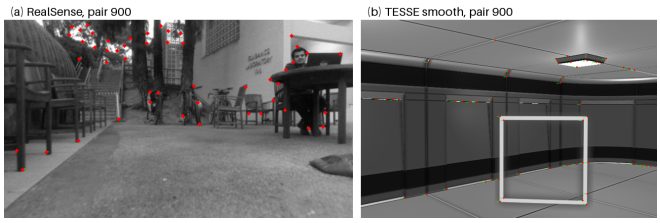


Fig. 5. Actual LK tracks at pair 900 of each bag. Green segments denote inter-image displacement and red points current locations. These are experiment outputs, not synthetic illustrations.

C. Geometric-estimation failure boundary

Version 004 rejected eight TESSE LK images after an OpenCV assertion inside `findFundamentalMat`. Version 005 retains the current image and valid LK correspondences, records the exception and exact points, and assigns an invalid geometric model with zero inliers. It neither relaxes thresholds nor substitutes another estimator.

Two saved finite, distinct correspondence sets of 11 and 12 points reproduce the assertion without ROS or image decoding. This localizes the failure to geometric estimation but does not establish its internal numerical cause. Final TESSE LK statistics contain 18 invalid geometric pairs, including insufficient-point/invalid-model cases; these rows remain in the means. Zero rejected images is therefore not a claim that every pair admits a valid F.

VI. SUPPLEMENTARY EXPERIMENTS AND LIMITS

A separate Python test on 30 EuRoC adjacent pairs compares SIFT, ORB, and FAST+BRIEF (Table IV). It is not a full official-bag comparison. ORB has a 500-point budget, while the other detectors produce many more points,

TABLE IV

SUPPLEMENTARY EUROC MEANS: PYTHON, 30 PAIRS

Method	Good matches	Inliers	Ratio
SIFT	1248.47	1178.93	94.15%
ORB	297.57	276.77	92.82%
FAST+BRIEF	3427.27	3140.67	91.24%

so counts and measured runtimes do not form a controlled speed/accuracy ranking. The supplementary Python SIFT box result also differs from the C++ regression and is not merged into Table I.

The Farneback dense-flow supplement uses a known synthetic translation $(2, 1)$ pixels. On 36,653 textured pixels, excluding a 30-pixel boundary, median endpoint error is 0.000291 pixels, the 95th percentile 0.003346 pixels, and the maximum 2.211763 pixels. Low central errors coexist with outliers and do not establish ground-truth accuracy on real images.

SURF is unavailable because the existing OpenCV build disables nonfree algorithms; the environment was not rebuilt. ORB, FAST+BRIEF, and dense flow remain separate Python supplements, not completed C++ derived trackers. Local grading determines whether these are optional; the broader MIT comparison requirements should not be declared entirely satisfied.

VII. REPRODUCIBILITY AND CONCLUSION

Final source version 005 is stored under the server's Lab5 version directory, with `course_entry.cpp` as the ROS entry. Full-frame evidence is in `lab5/results/official-005`, relative to the VNAV project root. That server version retains exact commands, checksums, and numerical-failure reproduction scripts. The accompanying report archive documents their locations. Reruns require new output paths.

The work establishes a complete SIFT/LK correspondence pipeline on both required recordings, supported by exact geometry proofs and exhaustive timestamp checks. Descriptor ambiguity, flow inconsistency, and geometric-model failure are treated separately. Keeping unsuccessful model estimates in the denominator is more informative than reporting only favorable frames. The demonstrated scope is feature matching and tracking, not complete SLAM or a real-time guarantee.

REFERENCES

[1] MIT VNAV, "Lab 5: Exercises," ROS1 2023. `https://vnav.mit.edu/labs_2023/lab5/exercises.html`, accessed Oct. 23, 2026.
[2] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.
[3] M. Muja and D. G. Lowe, "Fast approximate nearest neighbors with automatic algorithm configuration," in *VISAPP*, 2009.
[4] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *IJCAI*, pp. 674–679, 1981.
[5] C. Harris and M. Stephens, "A combined corner and edge detector," in *Alvey Vision Conf.*, pp. 147–151, 1988.
[6] OpenCV, "Feature detection and description; optical flow," OpenCV 4 documentation. `https://docs.opencv.org/4.x/`.